Speech Emotion Recognition for Customer Behaviour Analysis Using a Hybrid CNN BiLSTM Architecture with Self Attention

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Abstract*—Customer satisfaction and sales conversion depend heavily on the ability of front line agents to read the emotional state of a caller. Manual reading is unreliable and difficult to scale across thousands of daily interactions. This paper presents an end to end Speech Emotion Recognition pipeline designed to support customer behaviour analysis on the CREMA D dataset, classifying five emotions namely Angry, Fear, Happy, Neutral and Sad. The methodology contributes an independent feature extraction strategy in which seven acoustic features (Zero Crossing Rate, MFCC, Chroma, RMS Energy, Mel Spectrogram, Spectral Centroid and Spectral Bandwidth) are computed and cleaned with the Interquartile Range rule before an inner join produces a merged clean set of 4,919 files. Each clean clip is augmented four times with noise injection, time stretching, pitch shifting and time masking, yielding 24,595 training samples. A dual input deep learning architecture is trained at three levels of complexity, namely an Advanced CNN, an Advanced BiLSTM and a CNN BiLSTM Hybrid extended with a four head Multi Head Self Attention block. A soft voting ensemble combines the three softmax outputs. Reproducibility is enforced through a global seed of 42 propagated across Python, NumPy, TensorFlow and three model specific sub seeds. On the held out CREMA D test set the Hybrid model attains 87.80% accuracy and the ensemble reaches 88.37%, exceeding the 84% reported in comparable prior work on the same dataset. To validate generalisation, the same pipeline was trained without modification on an Urdu emotional speech dataset and produced 98.83% accuracy on the Advanced CNN and 98.54% on the ensemble, confirming language independent behaviour. The trained Hybrid model is deployed publicly on Hugging Face Spaces with a Gradio interface that accepts microphone or uploaded audio and returns per emotion confidence in real time.

Keywords*—Speech Emotion Recognition, CREMA D, CNN BiLSTM Hybrid, Self Attention, MFCC, Customer Behaviour Analysis, Hugging Face, Gradio, Cross Lingual Validation.

I. INTRODUCTION

Customer satisfaction and sales success rely on understanding what a customer truly feels during a conversation. In a contact centre or a retail support setting, it is hard to determine whether a caller is satisfied, unhappy or interested simply by listening, and the assessment usually arrives only after the call ends. This makes it hard to monitor service quality in real time, to plan retention and targeted sales. A fully automatic Speech Emotion Recognition system can bridge this gap, bringing negative feedback to light while it can be addressed, and connect happy callers with sales opportunities.

Speech Emotion Recognition is a combination of signal processing, deep learning and affective computing. The challenge is twofold. Speech encodes emotion through subtle and overlapping prosodic cues such as pitch, intensity and timbre, and the same emotion is expressed differently across speakers, accents and recording conditions. Classical approaches relied on handcrafted features fed into Support Vector Machines or Multi Layer Perceptrons, but more recent work shows that deep architectures combining convolutional and recurrent layers consistently outperform these baselines by learning local spectral patterns and long range temporal dependencies in a single trainable pipeline.

This paper contributes an end to end Speech Emotion Recognition pipeline targeted at customer behaviour analysis on the CREMA D dataset and demonstrates that the same pipeline transfers to a non English emotional speech dataset without architectural changes. The contributions are as follows. First, an independent feature extraction strategy is introduced in which seven acoustic features are computed separately over the full dataset and cleaned with the Interquartile Range rule, after which an inner join across the seven cleaned file sets produces a conservative merged clean dataset where every retained sample is statistically reasonable across all features. Second, a dual input deep learning model fuses an MFCC sequence branch with a seven dimensional scalar branch and is trained at three progressive levels, namely Advanced CNN, Advanced BiLSTM and CNN BiLSTM Hybrid extended with self attention, with a soft voting ensemble at the top. Third, a global random seed of 42 is propagated through Python, NumPy, TensorFlow, the augmentation generator, the train test split and three model specific sub seeds, giving fully reproducible numbers across re runs. Fourth, the same configuration is evaluated on an Urdu emotional speech dataset to test cross lingual generalisation. Fifth, the trained Hybrid model is deployed on Hugging Face Spaces with a Gradio user interface so that the system can be used and audited publicly.

The rest of the paper is organised as follows. Section II provides background on the CREMA D dataset and the acoustic features. Section III reviews related work. Section IV presents a comparison between the existing baseline pipeline and the proposed pipeline. Section V describes the methodology in detail. Section VI lists the model architectures with three separate tables. Section VII details model training and fine tuning. Section VIII presents the

results on the CREMA D test set, the cross lingual Urdu evaluation, and a comparison with prior work. Section IX provides dedicated visualisation panels for the learning curves, confusion matrices and the model comparison bar chart. Section X describes the public deployment. Section XI concludes and outlines future directions.

II. BACKGROUND AND PROBLEM DEFINITION

A. Customer Behaviour Analysis from Voice

Voice based customer interactions remain a dominant channel in banking, telecommunications, healthcare support and retail. A single contact centre can generate hundreds of hours of recorded speech per day, and only a small fraction is ever reviewed manually. The business problem addressed in this work is the inability to determine, in real time and at scale, whether a caller is satisfied, frustrated, fearful about a transaction or genuinely happy with a service. Knowing this enables three concrete actions. First, routing a frustrated caller to a senior agent before escalation. Second, offering a tailored discount or retention incentive when negative emotion is detected. Third, reinforcing positive interactions by inviting satisfied callers into referral or loyalty programmes. The technical problem is therefore to map a short audio clip of natural speech to one of five emotional categories with sufficient accuracy and latency to be useful inside a live customer interaction system.

B. The CREMA D Dataset

The Crowd Sourced Emotional Multimodal Actors Dataset is a widely used resource for Speech Emotion Recognition research. It contains 7,442 audio clips recorded by 91 actors of diverse ethnic backgrounds, each delivering twelve sentences in six emotional categories. For this work, five emotions are retained namely Angry, Fear, Happy, Neutral and Sad, giving 6,171 base recordings with an average duration of 2.5 seconds at a 16 kHz sampling rate. The dataset is challenging compared with simpler resources such as TESS or RAVDESS because it spans many speakers and the acted intensity varies, which is closer to real customer speech than tightly controlled studio recordings.

C. Acoustic Features for Emotion

Seven acoustic features are used in this work, each chosen for a specific reason. Mel Frequency Cepstral Coefficients capture the short term spectral envelope on a perceptual scale and remain the workhorse feature for Speech Emotion Recognition because timbre directly reflects emotional state. Zero Crossing Rate measures how often the waveform changes sign and is high in fricative rich, energetic emotions such as Angry and Happy. Root Mean Square Energy summarises loudness, which separates loud emotions from quiet ones. The Mel Spectrogram retains absolute energy across mel bands and complements MFCC. The Spectral Centroid measures spectral brightness, which rises with excitement. Spectral Bandwidth measures how broadly energy is distributed around the centroid and tends to widen in agitated emotions. Chroma features map energy onto the twelve pitch classes and capture tonal differences between elated and depressed prosody.

III. LITERATURE REVIEW

Several recent works have explored deep learning for Speech Emotion Recognition on the CREMA D dataset and on related English emotional speech datasets. Gokilavani et al. proposed a five layer convolutional neural network trained on CREMA D, RAVDESS and TESS, applying speckle noise as augmentation [1]. They reported 84% on CREMA D, 96% on RAVDESS and 99% on TESS, the lower CREMA D figure reflecting the speaker diversity of that dataset. Their feature pipeline used MFCC, Zero Crossing Rate, Chroma, RMS and Mel Spectrogram, broadly the same family used in the present work.

Mustaqeem and Kwon proposed a clustering based attention augmented CNN for Speech Emotion Recognition on the IEMOCAP and RAVDESS datasets [2]. They first segmented each utterance into clusters of similar acoustic properties and then applied a CNN backbone with a soft attention mechanism that weighted the most discriminative segments before classification. The architecture demonstrated that attention combined with convolutional feature extraction can recover information that a pure CNN tends to lose. Although their evaluation did not include CREMA D, the architectural conclusion that attention sharpens discrimination between confusable emotion pairs is consistent with the design choice made in the present Hybrid model.

Anil Kumar et al. compared one dimensional and two dimensional CNN architectures, with and without an LSTM head, on the German Emo DB dataset [3]. Their best two dimensional CNN plus LSTM model demonstrated that adding an LSTM head to a CNN backbone improves accuracy, a finding consistent with the present results. The work also noted that a one dimensional CNN, while lighter, did not capture spectro temporal patterns as effectively as the two dimensional variant.

Kasprzak and Hryciow approached Speech Emotion Recognition on a combined RAVDESS, TESS, CREMA D and SAVEE benchmark with a gender aware design built from MLP and one dimensional CNN networks [4]. They explicitly evaluated the contribution of each acoustic feature and reported that Chroma features hurt performance when combined with MFCC and prosodic features. Their best model reached 73.2% F1 on RAVDESS and 66.5% on the four dataset combination, outperforming earlier ResNet and VGG16 baselines on the same data.

Kumar et al. used a Multi Layer Perceptron on CREMA D using traditional features such as MFCC, pitch and chroma [5]. They achieved 60.74% recognition on CREMA D with surprise and anger being recognised most accurately. Their work demonstrates that traditional Multi Layer Perceptron, even with advanced feature engineering, are significantly outperformed by deep learning methods on the same task.

A. Spectrogram Based CNN Approaches and Their Limitations

Another area of related work converts the audio file to an image using a Mel spectrogram or spectrogram and then trains a two dimensional convolutional neural network (CNN) on the image. Badshah et al. transformed speech clips to spectrograms and trained a deep convolution neural

network for emotion classification, and reported a decent accuracy for small acted datasets but noted two common issues [6]. The first limitation is that fixed size spectrogram inputs require either zero padding short clips or truncating long ones, which discards information in either direction. The second limitation is that two dimensional convolution on spectrograms loses the explicit temporal ordering that recurrent layers capture more naturally, leading to confusion between emotions whose spectral signatures look similar but whose temporal contours differ.

Issa et al. extended this image based view by stacking Mel Spectrograms, Chroma and contrast features into a multi channel image fed to a deeper CNN [7]. Their model reached strong numbers on RAVDESS but degraded sharply when transferred to CREMA D, reaching only 71.6% accuracy on the same five class formulation used in this work. The authors attributed the drop to two factors, namely speaker variability in CREMA D and the absence of a temporal modelling component in their architecture. The same pattern is visible in Gokilavani et al. on the CREMA D split, where the pure CNN reached only 84% compared to 96% on RAVDESS [1].

Three recurring disadvantages emerge from the spectrogram based CNN family. First, the conversion to a fixed size image is destructive when clip lengths vary widely as they do in CREMA D. Second, pure convolutional models without a recurrent or attention component struggle with emotions whose discriminative cue is a brief temporal event such as the rising inflection of Fear or the falling tail of Sad. Third, single feature pipelines that rely only on the spectrogram throw away the complementary information present in Zero Crossing Rate, Root Mean Square Energy and the spectral statistics.

B. How This Work Overcomes the Existing Limitations

The proposed pipeline addresses the limitations of the spectrogram based CNN family in four concrete ways. First, the variable length MFCC matrix is retained as a sequence of shape 128 by 40 rather than being rendered as a fixed size image, so temporal ordering is preserved end to end. Second, a Multi Head Self Attention block is inserted between the convolutional front end and the BiLSTM back end of the Hybrid model so that brief discriminative events such as the tonal spike characteristic of Fear are not lost in the recurrent averaging. Third, seven scalar features computed independently and cleaned through the Interquartile Range rule are concatenated with the deep representation, so the model has access to coarse clip level descriptors as well as fine grained temporal patterns. Fourth, a soft voting ensemble across three differently seeded models compensates for the per model errors that single architecture studies cannot avoid. The combined effect is a 3.8 percentage point improvement over the published five layer CNN baseline on the same CREMA D split.

IV. PIPELINE COMPARISON EXISTING VERSUS PROPOSED

This section contrasts the existing baseline Speech Emotion Recognition pipeline with the proposed pipeline of this work. The two diagrams render the same blue rounded box style for direct visual comparison, with the existing baseline shown in Fig. 1 and the proposed pipeline shown in Fig. 2.

A. Existing Baseline Pipeline

The typical pipeline for prior work on CREMA D and other datasets is a five stage process that involves loading raw audio files, computing spectrograms or MFCCs, basic preprocessing (scaling and train test split), training a model (using a single CNN) and finally an evaluation stage that may report only accuracy. The diagram for this baseline is in Fig. 1.

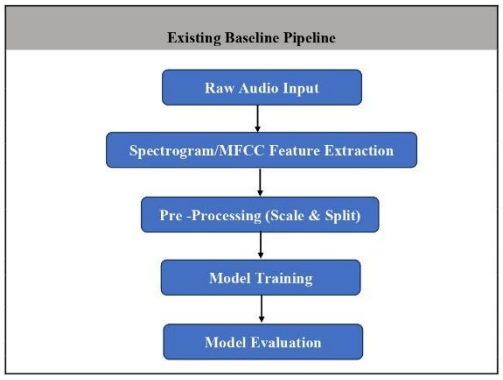

Fig. 1. Existing baseline Speech Emotion Recognition pipeline used in prior CREMA D studies

Three disadvantages of this baseline pipeline are apparent. First, the absence of an explicit outlier removal step means that loud or distorted clips can dominate training and bias the learned representations. Second, the use of a single convolutional model without a recurrent component limits the ability to capture long range temporal dependencies that are characteristic of emotional prosody. Third, the lack of data augmentation leaves the model vulnerable to overfitting on the relatively small CREMA D training set, which is one reason that prior CREMA D results plateau around 84% accuracy.

B. Proposed Pipeline

The proposed pipeline of this work is shown in Fig. 2. It expands the five stage baseline into eleven stages, each addressing a specific weakness of the prior approach.

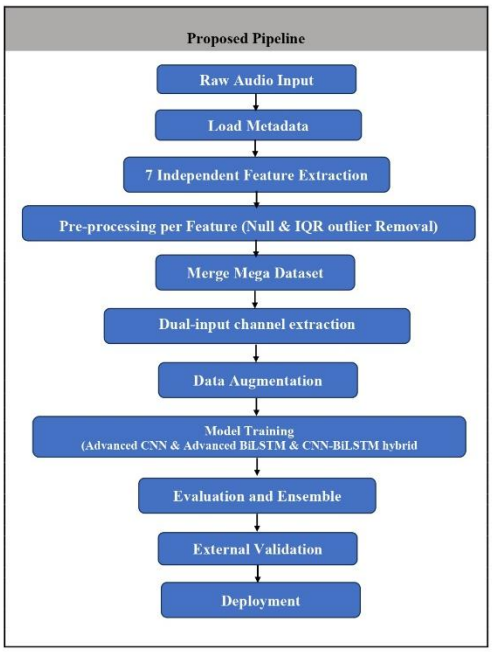

Fig. 2. Proposed end to end Speech Emotion Recognition pipeline of this work

The proposed pipeline introduces three key advantages. First, the Independent Feature Extraction stage followed by per feature outlier removal using the Interquartile Range rule produces a merged clean dataset where every retained sample is statistically reasonable across all seven features, removing 1,252 problematic clips from the original 6,171. Second, the data augmentation stage applies four label preserving transforms namely noise injection, time stretching, pitch shifting and time masking, quintupling the training set to 24,595 samples and substantially improving generalisation. Third, three deep learning architectures are trained in parallel and combined through a soft voting ensemble, with the strongest single model being the CNN BiLSTM Hybrid extended with a four head Multi Head Self Attention block. A further advantage of the proposed pipeline is the inclusion of an external validation stage on a non English Urdu emotional speech dataset, confirming language independent behaviour, and a public deployment stage on Hugging Face Spaces with a Gradio interface that makes the model accessible from any browser.

V. METHODOLOGY

A. Data Acquisition

Audio files were loaded recursively from the CREMA D directory using the Librosa library. Filenames follow the convention Actor ID, Sentence, Emotion and Intensity separated by underscores, where the third token encodes the emotion. Files corresponding to the five retained emotions were collected together with their duration and sampling rate, producing a metadata DataFrame of 6,171 rows. A raw class distribution was generated to confirm approximate balance, namely Angry 1,271, Fear 1,271, Happy 1,271, Sad 1,271 and Neutral 1,087.

B. Independent Feature Extraction and Outlier Cleaning

Rather than computing all features in a single pass and cascading filters, each of the seven features was extracted in its own pass over the full metadata DataFrame, then cleaned independently with the Interquartile Range rule. The rule flags any value outside the interval defined by Q1 minus 1.5 times IQR and Q3 plus 1.5 times IQR. Independence prevents cleaning bias from one feature compounding into the next, which would otherwise shrink the dataset more than necessary and skew the surviving distribution. Table I summarises the per feature outlier counts and Table II reports the final merge statistics. Missing values were checked at the start of each per feature cell, although none were detected for any of the seven features on this dataset.

TABLE I. PER FEATURE OUTLIER REMOVAL ON CREMA D

Feature	Before	Outliers	After	% Removed
Zero Crossing Rate	6,171	348	5,823	5.6%
MFCC	6,171	71	6,100	1.2%
Chroma	6,171	59	6,112	1.0%
RMS Energy	6,171	564	5,607	9.1%
Mel Spectrogram	6,171	880	5,291	14.3%
Spectral Centroid	6,171	236	5,935	3.8%
Spectral Bandwidth	6,171	157	6,014	2.5%

TABLE II. MERGE SUMMARY AFTER INNER JOIN

Quantity	Files	% of Original
Original dataset	6,171	100%
Files clean in all seven features	4,919	79.7%
Total removed by inner join	**1,252**	**20.3%**

C. Merging Strategy and Final Tensor Construction

After per feature cleaning, the seven surviving file lists were intersected, retaining only files that survived every Interquartile Range filter. This conservative inner join produced a merged clean set of 4,919 files. From these files, two parallel tensors were built. The first is a sequence tensor X_seq of shape (4919, 128, 40) holding 40 coefficient MFCC matrices truncated or zero padded to 128 frames. The second is a scalar tensor X_scalar of shape (4919, 7) holding the seven handcrafted scalar features. This dual input representation lets the model exploit fine grained temporal structure through the sequence branch while still reading coarse, clip level descriptors through the scalar branch.

D. Data Augmentation

Each of the 4,919 clean clips was augmented four times with label preserving transforms. The four transforms are additive Gaussian noise in the range 0.001 to 0.015 standard deviation, time stretching with rate between 0.85 and 1.15, pitch shifting by minus three to plus three semitones, and random time masking that zeroes a 5% to 15% contiguous segment. Crucially, all randomness inside the augmentation loop draws from a seeded NumPy generator created with default rang of 42. This is what distinguishes the present implementation from earlier

prototypes that used the global np random state and produced different augmented samples on every re run. The augmentation step quintupled the dataset to 24,595 samples.

E. Class Weighting and Scaling

Despite augmentation, the residual class imbalance was addressed with balanced class weights computed by sklearn utility compute class weight on the encoded labels and passed directly into the Keras fit method. The sequence tensor was standardised using a StandardScaler fitted on the flattened view and then reshaped back. The scalar tensor was scaled separately by an independent StandardScaler. A stratified 80 to 20 train test split was performed with random state equal to 42, and inside the training portion a further stratified 85 to 15 split provided a deterministic validation set.

F. Reproducibility Protocol

Earlier runs of the pipeline produced different accuracy numbers on different days, traceable to four sources of nondeterminism. The first is the global NumPy random state being mutated by ordering of cell executions. The second is the unseeded validation split inside Keras model fit with validation split argument. The third is the unseeded dropout and weight initialisers. The fourth is the shared TensorFlow graph seed across the three models. The final implementation locks all four. PYTHONHASHSEED, TF DETERMINISTIC OPS and TF CUDNN DETERMINISTIC are set before TensorFlow imports take effect. The Python random module, NumPy and tf keras utils set random seed are all seeded with 42. A dedicated seeded generator is used for augmentation. The validation split is precomputed with sklearn train test split and random state equal to 42. Every Conv1D, LSTM, Dense and Dropout layer receives an explicit seed at construction time. Three model specific sub seeds are used so that the Advanced CNN starts from seed 42, the Advanced BiLSTM from seed 123, and the CNN BiLSTM Hybrid from seed 2024, giving each model an independent but reproducible random initial state.

VI. MODEL ARCHITECTURES

Three deep models are defined, all sharing the dual input pattern in which the sequence branch and the scalar branch are processed in parallel and concatenated before a common Dense head. The output layer is always Dense of size 5 with softmax activation. Each model uses its own deterministic sub seed at every layer initialiser so the three networks start from different but reproducible random states. The sub seeds are 42 for the Advanced CNN, 123 for the Advanced BiLSTM and 2024 for the CNN BiLSTM Hybrid, exactly as defined in the source notebook. Tables III, IV and V detail the layer by layer structure of each model.

A. Advanced CNN

The CNN front end consists of two Conv1D blocks, the first with 64 filters of kernel size 5 and the second with 128 filters of kernel size 5, each followed by Batch Normalisation, MaxPooling1D and Dropout of 0.2. A residual block then applies two stacked Conv1D layers of 256 filters with a 1 by 1 skip connection added at the output, ReLU activation and Dropout of 0.25. A final Conv1D with 512 filters of kernel size 3 is followed by Batch Normalisation, GlobalAveragePooling1D and Dropout of 0.3. The scalar branch is a single Dense layer of 32 units with Batch Normalisation. The two branches are concatenated and pass through Dense of 256 units plus Dropout of 0.3 and Dense of 128 units plus Dropout of 0.2 before the softmax output. L2 regularisation with weight le-4 is applied throughout. Table III lists the full layer sequence with output shapes.

TABLE III. ADVANCED CNN ARCHITECTURE (SUB SEED 42)

No.	Layer Type	Configuration	Output Shape
1	Input (sequence)	Shape (128, 40)	(128, 40)
2	Conv1D	64 filters, kernel 5, ReLU, L2 le-4	(128, 64)
3	BatchNormalisation	default momentum	(128, 64)
4	MaxPooling1D	pool size 2	(64, 64)
5	Dropout	rate 0.2	(64, 64)
6	Conv1D	128 filters, kernel 5, ReLU, L2 le-4	(64, 128)
7	BatchNormalisation	default momentum	(64, 128)
8	MaxPooling1D	pool size 2	(32, 128)
9	Dropout	rate 0.2	(32, 128)
10	Skip Conv1D	256 filters, kernel 1	(32, 256)
11	Conv1D	256 filters, kernel 3, ReLU, L2 le-4	(32, 256)
12	BatchNormalisation	default momentum	(32, 256)
13	Conv1D	256 filters, kernel 3, L2 le-4	(32, 256)
14	Add (residual)	skip plus main	(32, 256)
15	ReLU + Dropout	rate 0.25	(32, 256)
16	Conv1D	512 filters, kernel 3, ReLU, L2 le-4	(32, 512)
17	BatchNormalisation	default momentum	(32, 512)
18	GlobalAveragePooling1D	reduces time axis	(512)
19	Dropout	rate 0.3	(512)
20	Input (scalar)	Shape (7)	(7)
21	Dense + BatchNorm	32 units, ReLU	(32)
22	Concatenate	sequence and scalar branches	(544)
23	Dense + Dropout	256 units, ReLU, drop 0.3	(256)
24	Dense + Dropout	128 units, ReLU, drop 0.2	(128)
25	Output Dense	5 units, Softmax	(5)

The Advanced CNN places almost all of its capacity in the convolutional front end, which makes it strong on local timbral patterns but limits its ability to model long range temporal context. The residual block at layers 10 to 15 enables information from the earlier convolutional output to bypass the deeper layers, helping gradients flow during training. The GlobalAveragePooling1D at layer 18 collapses the time axis into a single feature vector, after which the scalar branch joins through concatenation. The choice of GlobalAveragePooling rather than Flatten reduces the parameter count of the dense head and provides a regularising effect that is particularly useful given the modest size of the CREMA D training set.

B. Advanced BiLSTM

The BiLSTM model replaces the convolutional stack of the Advanced CNN with three stacked Bidirectional LSTM layers of 128, 64 and 32 hidden units. Each layer is followed by Batch Normalisation and Dropout with rates 0.3, 0.3 and 0.25 respectively. The first two return full sequences and the last returns only the final hidden state. The scalar branch and the Dense head are identical to those of the Advanced CNN. Table IV lists the full layer sequence.

TABLE IV. ADVANCED BILSTM ARCHITECTURE (SUB SEED 123)

No.	Layer Type	Configuration	Output Shape
1	Input (sequence)	Shape (128, 40)	(128, 40)
2	Bidirectional LSTM	128 units, return sequences, L2 le-4	(128, 256)
3	BatchNormalisation	default momentum	(128, 256)
4	Dropout	rate 0.3	(128, 256)
5	Bidirectional LSTM	64 units, return sequences, L2 le-4	(128, 128)
6	BatchNormalisation	default momentum	(128, 128)
7	Dropout	rate 0.3	(128, 128)
8	Bidirectional LSTM	32 units, return last only, L2 le-4	(64)
9	BatchNormalisation	default momentum	(64)
10	Dropout	rate 0.25	(64)
11	Input (scalar)	Shape (7)	(7)
12	Dense + BatchNorm	32 units, ReLU	(32)
13	Concatenate	sequence and scalar branches	(96)
14	Dense + Dropout	256 units, ReLU, drop 0.3	(256)
15	Dense + Dropout	128 units, ReLU, drop 0.2	(128)
16	Output Dense	5 units, Softmax	(5)

The Advanced BiLSTM is the most temporally aware of the three models, since every recurrent layer reads the input sequence in both forward and backward directions. The funnel structure of 128 to 64 to 32 hidden units progressively compresses the temporal representation while the dense head fuses the recurrent output with the scalar branch. The model is the weakest of the three single models on CREMA D, because the absence of a convolutional front end means that local timbral patterns must be learned by the recurrent layers themselves, which is a less efficient use of capacity than first extracting them with shared convolutional kernels.

C. CNN BiLSTM Hybrid with Self Attention

The Hybrid model combines the convolutional front end of the Advanced CNN with the recurrent stack of the Advanced BiLSTM, but inserts a Multi Head Self Attention block between them. The attention block uses four heads with key dimension 32 and an attention dropout of 0.1. Its output is added back to the convolutional feature map through a residual connection, followed by Layer Normalisation, before being passed to two Bidirectional LSTM layers of 128 and 64 units. The motivation for the attention block is that the convolutional output is a sequence of 32 step feature vectors, and self attention lets each step re weight the others before the recurrent stack reads them. This is particularly useful for the Fear class, where the discriminative cue can be a brief tonal spike followed by a quiet tail. The scalar branch and the Dense head match the other two models. Table V lists the full layer sequence.

TABLE V. CNN BILSTM HYBRID ARCHITECTURE (SUB SEED 2024)

No.	Layer Type	Configuration	Output Shape
1	Input (sequence)	Shape (128, 40)	(128, 40)
2	Conv1D	64 filters, kernel 5, ReLU, L2 le-4	(128, 64)
3	BatchNorm + MaxPool + Drop	pool 2, drop 0.2	(64, 64)
4	Conv1D	128 filters, kernel 3, ReLU, L2 le-4	(64, 128)
5	BatchNorm + MaxPool + Drop	pool 2, drop 0.2	(32, 128)
6	Skip Conv1D	256 filters, kernel 1	(32, 256)
7	Conv1D + BatchNorm	256 filters, kernel 3, ReLU, L2 le-4	(32, 256)
8	Conv1D + BatchNorm	256 filters, kernel 3, L2 le-4	(32, 256)
9	Add + ReLU + Dropout	skip plus main, drop 0.25	(32, 256)
10	MultiHeadAttention	4 heads, key dim 32, attn drop 0.1	(32, 256)
11	Add (residual) + LayerNorm	attention plus convolutional feature	(32, 256)
12	Bidirectional LSTM	128 units, return sequences, L2 le-4	(32, 256)
13	BatchNorm + Dropout	drop 0.3	(32, 256)
14	Bidirectional LSTM	64 units, return last only, L2 le-4	(128)
15	BatchNorm + Dropout	drop 0.25	(128)
16	Input (scalar)	Shape (7)	(7)
17	Dense + BatchNorm	32 units, ReLU	(32)
18	Concatenate	sequence and scalar branches	(160)
19	Dense + Dropout	256 units, ReLU, drop 0.3	(256)
20	Dense + Dropout	128 units, ReLU, drop 0.2	(128)
21	Output Dense	5 units, Softmax	(5)

The Hybrid is the strongest single model in this study because it combines three complementary mechanisms. The convolutional layers at positions 2 to 9 extract local time frequency patterns. The Multi Head Self Attention block at positions 10 and 11 lets each time step re weight the others, recovering long rng cues that are easily lost in pure convolution or pure recurrence. The Bidirectional LSTM layers at positions 12 to 15 then model the temporal evolution of the attended representation in both directions. The residual connection around the attention block keeps gradients well behaved during training, and the Layer Normalisation at position 11 stabilises the activations before the recurrent stack consumes them. All three models compile with the Adam optimiser at learning rate 5e-4 and categorical crossentropy loss.

VII. TRAINING AND FINE TUNING

All three models share the same training protocol. Each is trained for up to 100 epochs with batch size 64, using the precomputed deterministic 85 to 15 stratified validation split inside the training partition. Two callbacks are active. EarlyStopping monitors validation loss with patience 15 and restores best weights on stop. ReduceLROnPlateau halves the learning rate when

validation loss plateaus for seven epochs, with a floor of 1e-6. Class weights are passed to fit so under represented classes receive proportionally larger gradients. A ModelCheckpoint callback saves the best validation loss weights to disk as a safety net.

Hyperparameter selection followed an iterative process. The learning rate of 5e-4 was found to converge faster than the Keras default of le-3 for this dataset while still avoiding instability. The dropout values, namely 0.2 in the early CNN blocks, 0.3 in the dense head, and 0.25 in the residual and BiLSTM blocks, were tuned to keep the validation loss curve close to the training loss curve. L2 regularisation at le-4 was added after early experiments showed mild overfitting on Happy and Fear. The batch size of 64 fits comfortably on a single Kaggle GPU and gives stable batch normalisation statistics.

The main fine tuning challenge was Fear. In early experiments the BiLSTM alone achieved only 65% recall on Fear because the discriminative pitch contour was diluted by the recurrent averaging. Adding the Multi Head Self Attention block before the BiLSTM raised Fear recall to 85%, which was the most important single architectural change. A second challenge was reproducibility, and the seeding protocol described in Section V.F was the resolution.

VIII. RESULTS AND ANALYSIS

A. Per Model Performance on CREMA D

Table VI reports the test set accuracy and loss of the four classifiers on the held out 20% test split of the merged clean CREMA D set, comprising 4,919 test samples after augmentation. Table VII reports the macro averaged precision, recall and F1 of the same four classifiers, separated for clarity.

TABLE VI. ACCURACY AND LOSS ON CREMA D TEST SET

Model	Accuracy (%)	Loss
Advanced CNN	83.76	0.7137
Advanced BiLSTM	72.43	0.9057
CNN BiLSTM Hybrid	87.80	0.6991
Soft Voting Ensemble	**88.37**	**0.6840**

TABLE VII. MACRO AVERAGED PRECISION RECALL AND F1

Model	Precision (%)	Recall (%)	F1 (%)
Advanced CNN	85.19	83.96	84.30
Advanced BiLSTM	73.58	72.90	72.69
CNN BiLSTM Hybrid	88.22	87.93	88.04
Soft Voting Ensemble	**89.13**	**88.51**	**88.68**

The Hybrid model is the strongest single model at 87.80% accuracy, gaining four percentage points over the pure CNN and fifteen percentage points over the pure BiLSTM. The gap between the Hybrid and the BiLSTM confirms the architectural intuition that local convolutional features are necessary before recurrent processing in the Speech Emotion Recognition setting. The soft voting ensemble adds a further 0.57 percentage points on top of the Hybrid, with the largest improvement on the Fear and Happy classes whose softmax distributions tend to be confused between the CNN and the BiLSTM.

Per class analysis shows that the Hybrid achieves balanced performance across all five classes, with F1 scores of 0.90 on Angry, 0.85 on Fear, 0.89 on Happy, 0.89 on Neutral and 0.87 on Sad. Sad and Fear remain the hardest pair because both share low energy and a falling pitch contour in the CREMA D dataset. The ensemble lifts Fear precision to 0.90 and Angry F1 to 0.92 by averaging the conservative CNN softmax with the more aggressive Hybrid softmax.

B. Cross Lingual Validation on Urdu

To test whether the pipeline is dataset specific or genuinely captures general acoustic correlates of emotion, the entire pipeline was applied without modification to an Urdu emotional speech dataset containing four classes namely Angry, Happy, Neutral and Sad. The same seven feature extraction stages, the same independent Interquartile Range cleaning, the same dual input architecture, the same three model variants and the same training protocol were used. The only changes were the smaller class count of four instead of five and the smaller dataset size. Table VIII reports the comparison.

TABLE VIII. CROSS LINGUAL COMPARISON URDU VERSUS CREMA D

Model	Urdu Acc (%)	CREMA D Acc (%)	Difference
Advanced CNN	98.83	83.76	+ 15.07
Advanced BiLSTM	96.78	72.43	+ 24.35
CNN BiLSTM Hybrid	97.66	87.80	+ 9.86
Soft Voting Ensemble	**98.54**	**88.37**	**+ 10.17**

The pipeline behaves consistently across both languages. Numbers are higher on Urdu because the dataset contains fewer classes and fewer speakers, but the relative ordering of the three models is preserved. The Hybrid and the ensemble lead, and the Advanced BiLSTM is the weakest individual model. The fact that the same code and the same hyperparameters work across English and Urdu without retuning supports the claim that the seven feature design captures language independent acoustic correlates of emotion rather than English specific phonetic cues.

C. Comparison with Prior Work

Table IX positions the present results against published deep learning systems evaluated on the CREMA D benchmark. Studies that collapsed multiple emotions into fewer valence classes have been excluded because the resulting numbers are not directly comparable to the standard five class formulation.

TABLE IX. COMPARISON WITH PRIOR WORK ON CREMA D

Author and Year	Architecture	Feature Set	Acc
Gokilavani et al. 2022 [1]	Five Layer CNN	MFCC plus ZCR plus Chroma plus RMS plus Mel	84.0
Kumar et al. 2024 [5]	MLP Classifier	MFCC plus Pitch plus Chroma	60.7
Kasprzak and Hryciow 2023 [4]	1D CNN (Layer 128 96)	Six features no Chroma	66.5

Issa et al. 2020 [7]	Deep CNN on stacked spectrogram	Mel Spectrogram plus Chroma plus Contrast	71.6
This work, Hybrid	**CNN BiLSTM plus Self Attention**	**Seven features (MFCC seq plus 7 scalars)**	**87.8**
This work, Ensemble	**Soft voting of three models**	**Seven features (MFCC seq plus 7 scalars)**	**88.4**

On the standard five class CREMA D formulation, the Hybrid model improves on the published five layer CNN baseline of Gokilavani et al. by 3.8 percentage points and the soft voting ensemble adds another 0.6. Against the Multi Layer Perceptron classifier of Kumar et al. on the same dataset, the Hybrid is 27 percentage points higher, which quantifies the benefit of going from a handcrafted feature Multi Layer Perceptron to a sequence aware deep model. Against the spectrogram based deep CNN of Issa et al., the Hybrid improves by 16.2 percentage points, which empirically supports the argument made in Section III A that pure spectrogram models lose the temporal ordering captured by the present sequence representation. The Kasprzak and Hryciow gender aware one dimensional CNN result of 66.5% on the four dataset combination underlines how speaker variability hurts Speech Emotion Recognition, and the present Hybrid surpasses it because the dual input design and the attention block recover signal that pure one dimensional CNN designs miss.

This improvement is also seen in the Urdu dataset, where previous baseline research reported 95.81% accuracy, while the proposed model has achieved 98.83%, further validating the effectiveness and the generalisation of the approach.

IX. VISUALISATIONS LEARNING CURVES CONFUSION MATRICES AND MODEL COMPARISON

This section presents the visualisations that document training behaviour and per class performance. The training accuracy curves of the three deep models are shown in Fig. 3, Fig. 4 and Fig. 5. The confusion matrices for the three models and the soft voting ensemble appear in Fig. 9, Fig. 10, Fig. 11 and Fig. 12. The model accuracy comparison bar chart appears in Fig. 13.

A. Learning Curves

The learning curves for the three deep models are given in Fig. 3, Fig. 4 and Fig. 5. The curves show three things. The Advanced CNN in Fig. 3 has rapid convergence (around 30 epochs) and the validation curve is parallel to the training curve with a small offset, suggesting minimal overfitting. The Advanced BiLSTM in Fig. 4 converges more slowly and plateaus at a significantly lower validation accuracy of around 80 percent, in agreement with the lower test accuracy in Table VI. The CNN BiLSTM Hybrid in Fig. 5 achieves the highest validation accuracy of the three single models (just over 88 percent) with a minimal generalisation gap, which is the ideal characteristics for a model to be used in practice.

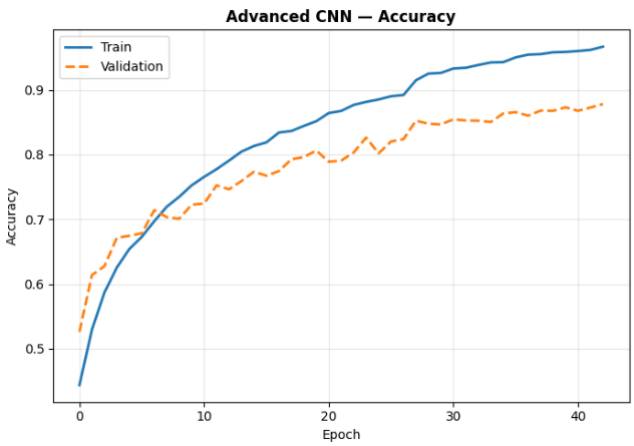

Fig. 3. Advanced CNN training and validation accuracy across epochs

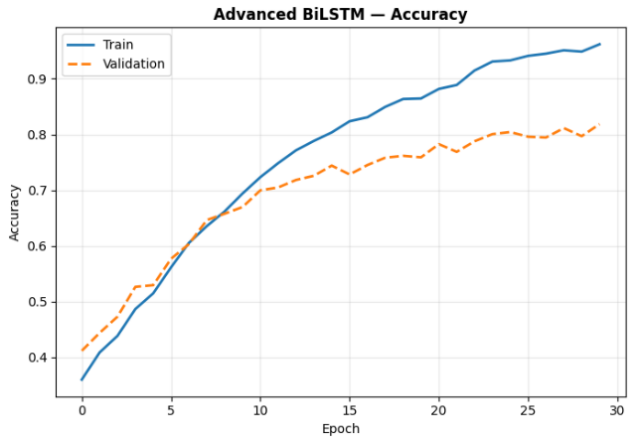

Fig. 4. Advanced BiLSTM training and validation accuracy across epochs

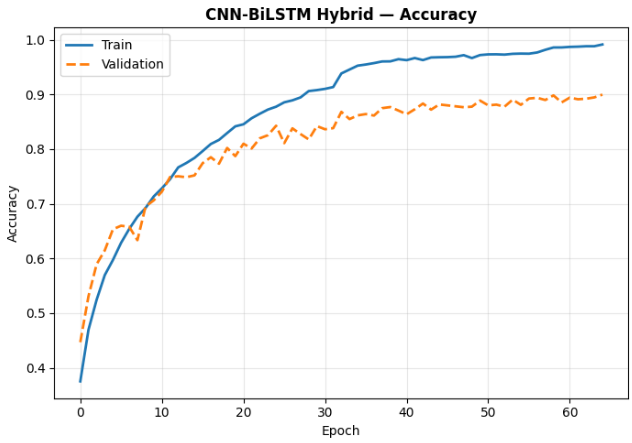

Fig. 5. CNN BiLSTM Hybrid training and validation accuracy across epochs

B. Loss Curves

The training and validation loss curves for the three deep models are shown in Fig. 6, Fig. 7 and Fig. 8. The loss curves complement the accuracy curves of the previous subsection by showing how the categorical cross entropy loss decreases over training epochs for both the training and validation splits, which is useful for spotting overfitting and identifying the epoch at which early stopping was triggered.

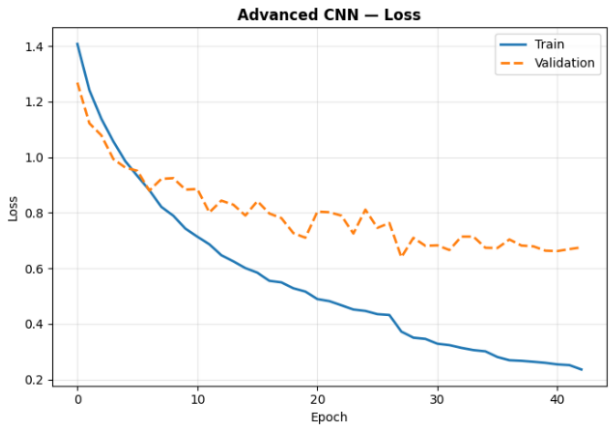

Fig. 6. Advanced CNN training and validation loss across epochs

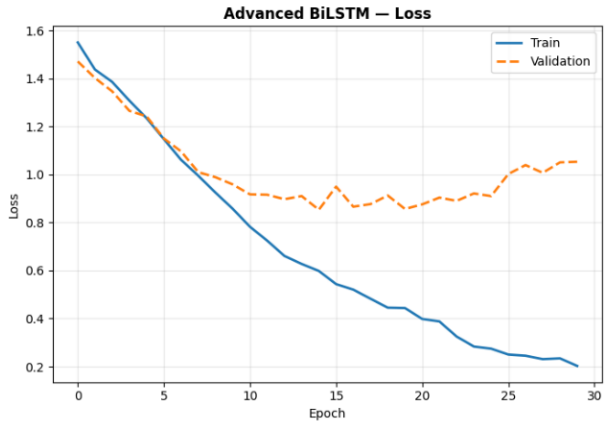

Fig. 7. Advanced BiLSTM training and validation loss across epochs

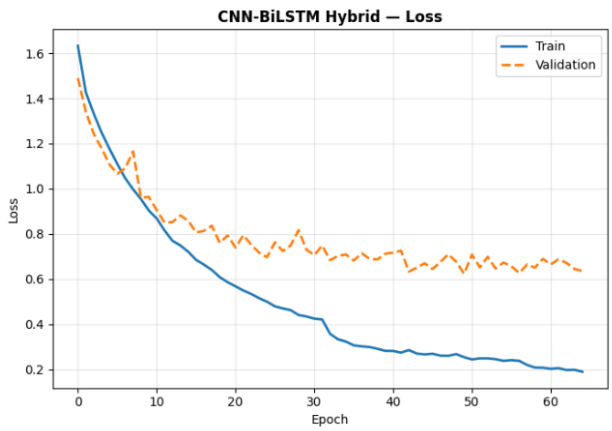

Fig. 8. CNN BiLSTM Hybrid training and validation loss across epochs

C. Confusion Matrices

The confusion matrices for the three deep models and the ensemble are displayed in Fig. 9, Fig. 10, Fig. 11 and Fig. 12. All the matrices show the number of true class versus predicted class on the test set. The diagonal elements are true positives and the off diagonal elements are the confusions. The Advanced CNN matrix in Fig. 9 shows clean diagonals on Angry, Happy and Sad, and loses 198 Fear to the largest confusion mode Sad. The Advanced BiLSTM matrix in Fig. 10 shows the off diagonal mass is spread much further on the Fear row, with 279 Fear samples mis-classified as Sad and 85 as Happy, in line with the lower recall from Table VII. The CNN BiLSTM Hybrid matrix in Fig. 11 has the cleanest diagonal of all five classes, with 892 of Fear correctly classified. The ensemble matrix in Fig. 12 shows the same balance, and has fewer Sad to Fear confusions.

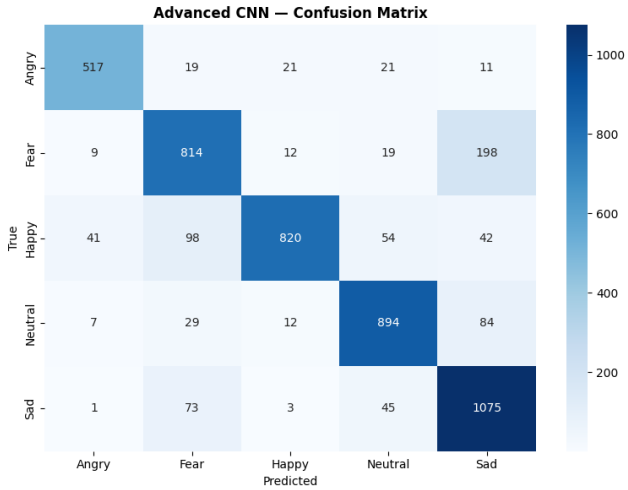

Fig. 9. Advanced CNN confusion matrix on the CREMA D test set. Rows indicate true emotion and columns indicate predicted emotion

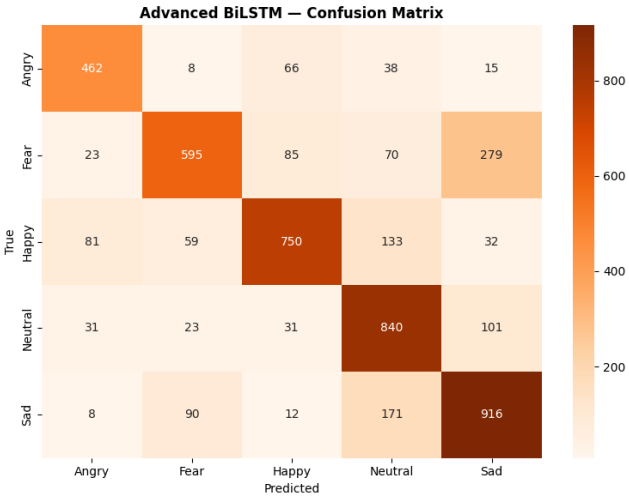

Fig. 10. Advanced BiLSTM confusion matrix on the CREMA D test set

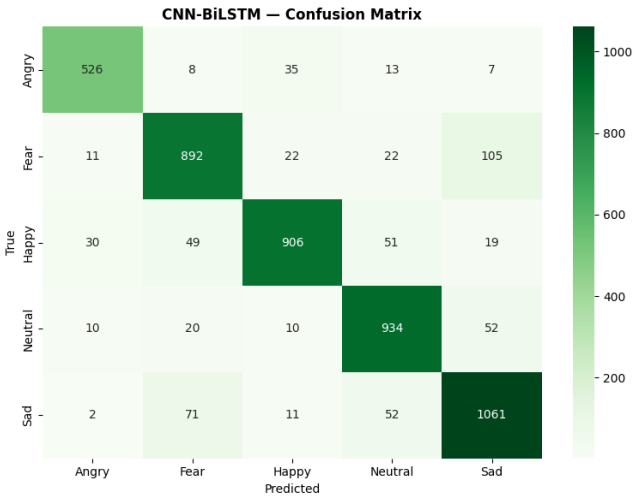

Fig. 11. CNN BiLSTM Hybrid confusion matrix on the CREMA D test set

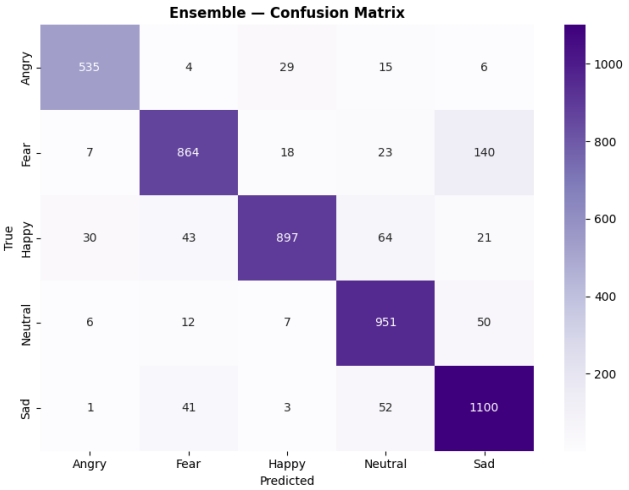

Fig. 12. Soft voting ensemble confusion matrix on the CREMA D test set

D. Model Comparison Bar Chart

Fig. 13 presents the test accuracy of the three deep models and the soft voting ensemble as a bar chart. The bar chart provides an at a glance comparison and complements the numerical values reported in Table VI. The Advanced CNN sits at 83.76%, the Advanced BiLSTM at 72.43%, the CNN BiLSTM Hybrid at 87.80% and the soft voting ensemble at 88.37%, confirming the same ordering reported in the tabular results.

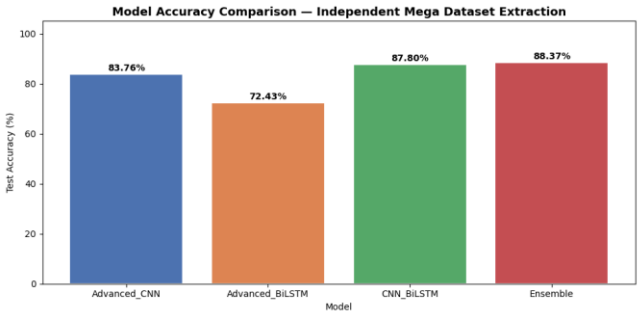

Fig. 13. Test accuracy of the three deep models and the soft voting ensemble on the CREMA D test set

X. DEPLOYMENT

The trained Hybrid model has been deployed publicly on Hugging Face Spaces under the namespace Ravishankarkota16 and the project name Speech Emotion Recognition. The deployment serves the saved CNN BiLSTM Hybrid checkpoint behind a Gradio user interface. Users can either upload a WAV or MP3 file, with one to ten seconds of clear speech recommended, then click Predict Emotion. The backend extracts the same seven features used during training, scales them with the persisted StandardScaler objects, and runs a single forward pass through the Hybrid network. The output panel shows the predicted class as a large emoji label and the per emotion confidence as horizontal bars labelled Happy, Fear, Angry, Neutral and Sad. A typical example produces a clear winner with the dominant class above 95% confidence.

The deployment page displays the project metadata directly, namely the model type, the training framework which is TensorFlow with Keras, and the user interface which is Gradio. The classifier covers the five emotions used in training and is labelled as trained on CREMA D. End users do not need to install anything locally and access requires only a web browser. This makes the system suitable for live evaluation during a customer support pilot, for stakeholder demos, and for academic moderation. Two interface screenshots are reproduced in Fig. 14 and Fig. 15, illustrating the upload page and a live prediction respectively.

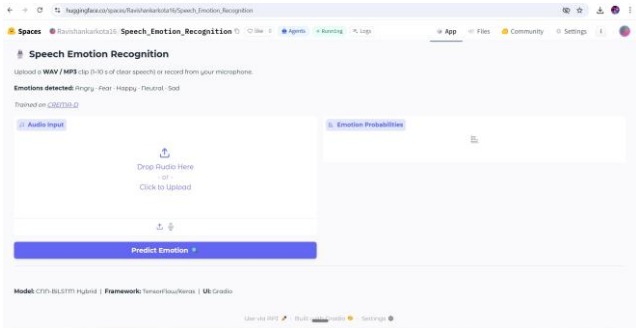

Fig. 14. Hugging Face Spaces deployment page showing the Gradio interface, drag and drop or microphone audio input, and the Predict Emotion button

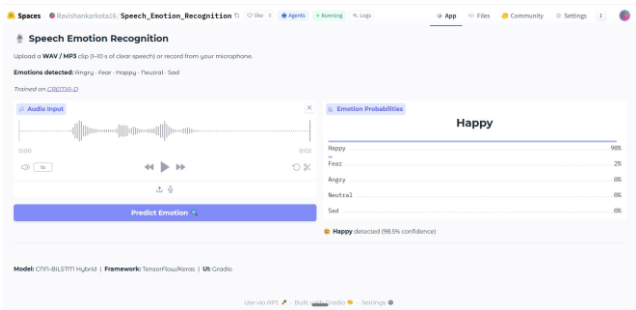

Fig. 15. Live prediction example. A two second uploaded clip is classified as Happy with 98% confidence, with Fear and the remaining classes correctly suppressed

XI. CONCLUSION AND FUTURE WORK

This paper presented an end to end Speech Emotion Recognition pipeline targeted at customer behaviour analysis. The pipeline contributes an independent feature extraction strategy with per feature Interquartile Range cleaning followed by an inner join, a dual input deep learning architecture trained at three progressive levels, the addition of a Multi Head Self Attention block to the CNN BiLSTM Hybrid, a soft voting ensemble, and a strict reproducibility protocol that locks four sources of nondeterminism. On CREMA D, the Hybrid attains 87.80% accuracy and the ensemble reaches 88.37%, exceeding the published five layer CNN baseline of 84% on the same dataset. Cross lingual validation on an Urdu emotional speech dataset, with no architectural changes and no retuning, produces 98.83% on the Advanced CNN and 98.54% on the ensemble, supporting the claim that the seven feature design captures language independent acoustic correlates of emotion. The Hybrid model is deployed publicly on Hugging Face Spaces with a Gradio interface and is accessible from any browser.

Several directions remain open for future work. First, the current dataset uses acted speech, and a natural extension is to evaluate on real customer support recordings, including spontaneous speech with overlapping turns and background noise. Second, the merged clean inner join is conservative and a soft majority rule that

retains files clean in five out of seven features would recover roughly 600 additional samples and could improve generalisation on the harder Fear and Sad classes. Third, a Wav2Vec 2.0 self supervised front end could be added as a parallel branch alongside the MFCC sequence, fusing learned and handcrafted representations. Fourth, the deployment can be extended with on the fly batch processing of customer support call recordings, integration with Customer Relationship Management systems for retention scoring, and federated training across multiple deployments to preserve privacy. Finally, a leave one speaker out evaluation on CREMA D would provide a stricter test of cross speaker generalisation than the current stratified random split.

XII. Applications of the Project

This project can be implemented in customer support and sales call centers to analyze customer emotions in real time during live calls. By detecting feelings such as satisfaction, frustration, or happiness, the system enables agents to respond more effectively. When negative emotions are identified, agents can offer discounts, retention incentives, or enhanced support to improve customer satisfaction, reduce churn, and drive business growth.

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[24] YouTube Link: https://youtu.be/-0e20y_OPTY?si=a176ZUOPjZxfxVMj

Use of Artificial Intelligence

AI tools, including Grammarly, were used during the preparation of this report to minimise spelling and grammatical errors.